**CAMBRIDGE INTERNATIONAL EXAMINATIONS**

**GCE Ordinary Level**

## MARK SCHEME for the May/June 2014 series

# 5070 CHEMISTRY

**5070/41** Paper 4 (Alternative to Practical), maximum raw mark 60

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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**1** **(a)** **(i)** $2Cu + O_2 \rightarrow 2CuO$ (1) [1]

**(ii)** black (1) [1]

**(b)** **(i)** 72 (1) $cm^3$ [1]

**(ii)** nitrogen (1) [1]

**(iii)** 18 (1) $cm^3$ [1]

**(iv)** 0.00075 (1) moles [1]

**(v)** 0.096 (1) g [1]

**(c)** 300 (1) $cm^3$ [1]

**[Total: 8]**

**2** **(a)** **(i)** red / pink (1) [1]

**(ii)** hydrochloric acid (1) [1]

**(iii)** Universal indicator / pH meter / full range indicator (1) [1]

**(b)** **(i)** diffusion (1) [1]

**(ii)** ammonium chloride **AND** $NH_4Cl$ (1) [1]

**(iii)** **C** (1)

Explanation
Ammonia molecules move or diffuse faster (than $HCl$ molecules), or reverse (1)

Ammonia has lower density <u>than $HCl$</u> / lower $M_r$ <u>than $HCl$</u> / ammonia molecules are lighter <u>than $HCl$ molecules</u>, or reverse (1)

If density of gases are compared to air, both densities must be stated e.g. ammonia lighter than air AND hydrogen chloride heavier than air. [3]

**(c)** **Y** ($NH_3$) (1); **X** ($HCl$) (1)

<u>Both</u> soluble in water (1)
$HCl$ is more dense than air **AND** $NH_3$ is less dense than air (1) [4]

**[Total: 12]**

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**3** **(d)** **[Total: 1]**

**4** **(b)** **[Total: 1]**

**5** **(a)** **[Total: 1]**

**6** **(b)** **[Total: 1]**

**7** **(a)** 1.70 (1) g [1]

**(b)** carbon dioxide (1) turns lime water milky / white ppt (1) [2]

**(c)** pink / red to yellow (1) [1]

**(d)**

| | | | | |
|---|---|---|---|---|
| 25.9 | 48.6 | 32.4 | (1) | 1 mark for each correct |
| 0.0 | 23.3 | 6.9 | (1) | row or column to the benefit of the candidate (3) |
| 25.9 | 25.3 | 25.5 | (1) | |

Mean value 25.4 (1) $cm^3$ [4]

**(e)** 0.00254 (1) moles [1]

**(f)** 0.00254 (1) moles [1]

**(g)** 0.0254 (1) moles [1]

**(h)** 0.05 (1) moles [1]

**(i)** 0.0246 (1) moles [1]

**(j)** 0.0123 (1) moles [1]

**(k)** 138 (1) 39 (1) [2]

**[Total: 16]**

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**8** **(a)** Transition metal ion / compound may be present (1)

**(b)** **(i)** green precipitate (1)

**(ii)** precipitate insoluble (1)

**(iii)** **gas** evolved that turns damp red litmus blue (1)
ammonia (1)

**(c)** $BaCl_2$ or $Ba(NO_3)_2$ or names(1)
$HCl$ or $HNO_3$ or names(1)
white ppt (1)

**[Total: 8]**

**9** **(a)** yellow (1) [1]

**(b)** 0.64, 1.27, 1.91, 2.35, 2.35, 2.35 all correct (1) [1]

**(c)** all points plotted correctly (1)
two straight lines, one of which must go through zero (1)
lines intersect (1) [3]

**(d)** **(i)** 3.2 (1) $cm^3$ [1]

**(ii)** 2.35 (1) g [1]

**(iii)** 7.4 (1) $cm^3$ [1]

**All answers in (d) must come from the candidate's graph. Read candidate's graph to +/– half a small square.**

**(e)** $AgNO_3 + KI \rightarrow AgI + KNO_3$ (1) [1]

**(f)** 1.35 (1) mol / $dm^3$ [1]

**(g)** $M_r$ $AgCl$, 143.5 (1)
Mass of $AgCl$ = 1.435 (1) g [2]

**[Total: 12]**

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